

Félix Gélinas

Toronto, Ontario, Canada

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Education

York University (Advisors: Nantel Bergeron and Vincent Pilaud) <i>Doctor of Philosophy in Mathematics (PhD)</i>	2023 – expected 2027 Toronto, Ontario, Canada
University of Toronto (Advisor: Swastik Kopparty) <i>Master of Science in Mathematics (MSc)</i>	2022 – 2023 Toronto, Ontario, Canada
Université du Québec à Montréal (UQAM) <i>Bachelor of Science in pure mathematics (BSc)</i>	2019 – 2022 Montréal, Québec, Canada

Research Experience

PhD project (Advisors: Nantel Bergeron and Vincent Pilaud) <i>York University</i>	September, 2023 – Present Toronto, Ontario, Canada
Supervision of Summer internship in Research Groups (York) <i>York University</i>	May, 2024 – August, 2024 Toronto, Ontario, Canada
Group Working On Hopf Algebra • Working with Mike Zabrocki, Lucas Gagnon • Supervising students Eric Schlums and Nicolas Andrews • Writing of an article • Programming (SageMath) • Hopf Algebra	
Group Working On Machine Learning • Working with Mike Zabrocki, Lucas Gagnon • Supervising students Eniola Ogunjimi and Inderpreet Ganda • Writing of an article • Programming (SageMath) • Machine Learning, Combinatorics	
Master Thesis project (Advisor: Swastik Kopparty (UofT)) <i>University of Toronto (UofT)</i>	May, 2023 – August, 2023 Toronto, Ontario, Canada
• Summarizing articles • Writing of a rapport of articles • Reading articles • Combinatorics, Graph Theory	
Summer internship in Research (Advisor: Hugh Thomas (UQAM)) <i>Université du Québec à Montréal (UQAM)</i>	May, 2022 – August, 2022 Montréal, Québec, Canada
• Writing of an article • Programming (SageMath) • Combinatorics, Dyck path	
Summer internship in Research (Advisor: Hugh Thomas (UQAM)) <i>Université du Québec à Montréal (UQAM)</i>	May, 2021 – August, 2021 Montréal, Québec, Canada
• Writing of an article • Programming (SageMath) • Algebra, Group theory	

Teaching Assistant (TA)

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York University

- Introduction to Sets and Logic (MATH1190)
- Discrete Mathematics for Computer Science (MATH1019)

January 2026 – Present

Toronto, Ontario, Canada

Teaching Assistant (TA)

York University

- Applied Calculus I (MATH1013)
- Discrete Mathematics for Computer Science (MATH1019)

September 2025 – December 2025

Toronto, Ontario, Canada

Teaching Assistant (TA)

York University

- Introduction to Logic for Computer Science (MATH1090)

May 2025 – August 2025

Toronto, Ontario, Canada

Teaching Assistant (TA)

York University

- Introduction to Logic for Computer Science (MATH1090)
- Problems, Conjectures and Proofs (MATH1200)

January 2025 – April 2025

Toronto, Ontario, Canada

Teaching Assistant (TA)

York University

- Differential Calculus with Applications (MATH1300)
- Introduction to Logic for Computer Science (MATH1090)
- Problems, Conjectures and Proofs (MATH1200)

September 2024 – December 2024

Toronto, Ontario, Canada

Teaching Assistant (TA)

York University

- Applied Multivariate and Vector Calculus (MATH2015)

May 2024 – June 2024

Toronto, Ontario, Canada

Teaching Assistant (TA)

York University

- Integral Calculus with Applications (MATH1310)
- Problems, Conjectures and Proofs (MATH1200)

January 2024 – April 2024

Toronto, Ontario, Canada

Teaching Assistant (TA)

York University

- Problems, Conjectures and Proofs (MATH1200)
- Differential Calculus with Applications (MATH1300)

September 2023, December 2023

Toronto, Ontario, Canada

Teaching Assistant (TA)

University of Toronto

- Calculus I (MAT135)
- Calculus II (MAT136)

September, 2022 – April 2023

Toronto, Ontario, Canada

Teaching Assistant (TA)

Université du Québec à Montréal (UQAM)

- Calculus of Differential Forms (MAT2410)

January, 2022 – April 2022

Montréal, Québec, Canada

Organization

Co-organizer for Participants-Seminar of the CGGC

Universitat de Barcelona (CRM)

- Co-organizer for Participants-Seminar of the CGGC Research School (Weekly from October 23 to November 20, 2025)

October, 2025 – November 2025

Barcelona, Spain

Other Experience

Moderator for Mathematical Summer in Paris

July, 2021 – July 2021

Université Paris-Saclay

Paris, France

- Moderator for an online mathematical conference (from July 5 to 9, 2021)

Preprint

Source Characterization of the Hypergraphic Posets. (<https://arxiv.org/abs/2508.16006>) – Félix Gélinas.

Ornamentation Lattices and Intreeval Hypergraphic Lattices. (<https://arxiv.org/abs/2508.01606>) – Antoine Abram, Jose Bastidas, Félix Gélinas, Vincent Pilaud, Andrew Sack.

When are Hopf Algebras Determined by Integer Sequences? (<https://arxiv.org/abs/2505.06941>) – Nicolas Andrews, Lucas Gagnon, Félix Gélinas, Eric Schlums, Mike Zabrocki.

Proof of a conjecture of Matherne, Morales, and Selover on encodings of unit interval orders (<https://arxiv.org/abs/2212.12171>) – Félix Gélinas, Adrien Segovia, Hugh Thomas. Accepted in The Electronic Journal of Combinatorics.

Publications

Gabriel Frieden, Félix Gélinas, and Étienne Soucy. “Cones and ping-pong in three dimensions”. In: *Involve, a Journal of Mathematics* 17.1 (Mar. 2024), pp. 11–28. ISSN: 1944-4176. DOI: 10.2140/involve.2024.17.11. URL: <http://dx.doi.org/10.2140/involve.2024.17.11>

Presentations (Talks)

Séminaires de mathématiques discrètes

December, 2025

Université de Liège

Liège, Liège, Belgique

- Polytopes et posets hypergraphiques
- Seminar Program

CGGC25

November, 2025

Universitat Autònoma de Barcelona

Barcelona, Catalunya, Spain

- Exploring Hypergraphic Polytopes

CGGC25 Participants Seminar

October, 2025

Universitat Autònoma de Barcelona

Barcelona, Catalunya, Spain

- Hypergraphic Polytopes and Posets

Canadian Discrete and Algorithmic Mathematics Conference (CanaDAM)

May, 2025

University of Ottawa

Ottawa, Ontario, Canada

- Source Characterization of the Hypergraphic Posets
- Conference Program

Algebraic and enumerative combinatorics seminar

May, 2025

University of Waterloo

Waterloo, Ontario, Canada

- Source Characterization of the Hypergraphic Posets
- Seminar Program

Winter School and Workshop on Polytopes*Universität Osnabrück*

- Source Characterization of the Hypergraphic Posets
- Conference Program

February, 2025

*Osnabrück, Lower Saxony, Germany***Séminaire du LACIM***Université du Québec à Montréal (UQAM)*

- Caractérisation du poset hypergraphique à partir des sources d'orientations acycliques
- Seminar Program

February, 2025

*Montréal, Québec, Canada***Combinatorial Algebra meets Algebraic Combinatorics (CAAC)***The Fields Institute*

- Source Characterization of the Hypergraphic Posets
- CAAC Program

January, 2025

*Toronto, Ontario, Canada***The Algebraic Combinatorics Workgroup***Nantel Bergeron (York University) – Vasu Tewari (UofT)*

- Hypergraphic polytopes | Part 1 - Part 2
- Part 1 : www.fields.utoronto.ca/talks
- Part 2 : www.fields.utoronto.ca/talks

October-November, 2024

*The Fields Institute, Toronto, Ontario, Canada***The Algebraic Combinatorics Workgroup***Nantel Bergeron– Kelvin Chan– Blessing Oni (York University)*

- Gently Discovering Gentle Algebras
- <https://garsia.math.yorku.ca/fieldseminar>
- Link to the presentation (YouTube)

May, 2024

*The Fields Institute, Toronto, Ontario, Canada***Graduate student Group***Leo Jiang (UofT)*

- Permutrees

May, 2024

*The Fields Institute, Toronto, Ontario, Canada***The Algebraic Combinatorics Workgroup***Nantel Bergeron– Kelvin Chan– Blessing Oni (York University)*

- The volume polynomial of a Matroid
- <https://garsia.math.yorku.ca/fieldseminar>
- Link to the presentation (YouTube)

September, 2023

*The Fields Institute, Toronto, Ontario, Canada***The Graduate Student Seminar UofT***Mathematics Departement UofT*

- Proof of a conjecture of Matherne, Morales, and Selover on encodings of unit interval orders

March, 2023

*Toronto, Ontario, Canada***The Algebraic Combinatorics Workgroup***Nantel Bergeron (York University)– Kelvin Chan (York University)*

- Cones and ping-pong in three dimensions
- <https://garsia.math.yorku.ca/fieldseminar>
- Link to the presentation (YouTube)

March, 2023

*The Fields Institute, Toronto, Ontario, Canada***Applied Algebra Seminar***Lucas Gagnon (York University)*

- Proof of a conjecture of Matherne, Morales, and Selover on encodings of unit interval orders
- <https://garsia.math.yorku.ca/seminar/algebra.html>

March, 2023

*York, Ontario, Canada***The Graduate Students Seminar (UQAM)***Université du Québec à Montréal (UQAM)*

- Cones and ping-pong in three dimensions

August, 2022

*Montréal, Québec, Canada***Pan-Québec colloquium for students of the ISM***Institute of Mathematical Sciences (ISM)*

- Cones and ping-pong in three dimensions
- <http://ism.uqam.ca/Colloque-panquebecois>

June, 2022

Québec, Québec, Canada

Presentations (Poster)

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| Combinatorial Algebra meets Algebraic Combinatorics (CAAC)
<i>Dalhousie University</i> <ul style="list-style-type: none">• Ornamentation Lattices and Intreeval Hypergraphic Lattices | January, 2026
<i>Halifax, Nova Scotia, Canada</i> |
| Au-delà du Permutoèdre et de l'associaèdre
<i>CIRM (Centre International De Rencontres Mathématiques)</i> <ul style="list-style-type: none">• Source Characterization of the Hypergraphic Posets | December, 2025
<i>Marseille, Provence-Alpes-Côte d'Azur, France</i> |
| CGGC25 Closing Conference
<i>Universitat Autònoma de Barcelona</i> <ul style="list-style-type: none">• Source Characterization of the Hypergraphic Posets• CGGC2025 Program | November, 2025
<i>Barcelona, Catalunya, Spain</i> |
| AlCoVe : an Algebraic Combinatorics Virtual Expedition <ul style="list-style-type: none">• Source Characterization of the Hypergraphic Posets• AlCoVe Program | May, 2025
<i>Online Conference</i> |
| VIII Encuentro Colombiano de Combinatoria (ECCO)
<i>Universidad del Cauca</i> <ul style="list-style-type: none">• Proof of a conjecture of Matherne, Morales, and Selover on encodings of unit interval orders• ECCO Program | June, 2024
<i>Popayán, Cauca, Colombia</i> |
| Combinatorial Algebra meets Algebraic Combinatorics (CAAC)
<i>Université du Québec à Montréal</i> <ul style="list-style-type: none">• The volume polynomial of a Matroid• CAAC Program | January, 2024
<i>Montréal, Québec, Canada</i> |
| Combinatorial Algebra meets Algebraic Combinatorics (CAAC)
<i>University of Waterloo</i> <ul style="list-style-type: none">• Proof of a conjecture of Matherne, Morales, and Selover on encodings of unit interval orders• CAAC Program | January, 2023
<i>Waterloo, Ontario, Canada</i> |

Awards & Honors

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| Academic Excellence Fund
<i>York University</i> | 2025 |
| FGS Tuition Fee Bursary
<i>York University</i> | 2025 |
| Kingsley Abu Imonikhe Graduate Award
<i>York University</i> | 2025 |
| Academic Excellence Fund
<i>York University</i> | 2025 |
| Vernon Oliver Stong Graduate Scholarships in Science
<i>York University</i> | 2024 |
| Graduate Studies
<i>York University</i> | 2023-2027 |
| Graduate Studies
<i>York University</i> | 2023-2024 |
| Graduate Studies
<i>University of Toronto</i> | 2022-2023 |

Undergraduate Student Research Awards <i>NSERC</i>	2022
Advisor contribution NSERC <i>Hugh Thomas</i>	2022
Supplements of the NSERC Undergraduate Student Research Awards <i>FRQNT</i>	2022
Undergraduate summer scholarship <i>Institute of Mathematical Sciences (ISM)</i>	2021
Advisor contribution ISM <i>Hugh Thomas</i>	2021
Advisor contribution ISM <i>Christophe Reutenauer</i>	2021
<i>Summer Schools</i>	
VIII Encuentro Colombiano de Combinatoria (ECCO) <i>CIMPA Research School, Popayán, Cauca, Colombia</i>	2024
ISM Discovery School: Advances in Geometric Combinatorics and Representation Theory <i>R. González D’Léon – A. H. Morales – F. Saliola – H. Thomas, Montréal, Québec, Canada</i>	2024
<i>Winter Schools</i>	
Winter School and Workshop on Polytopes <i>M. Juhnke – M. Reitzner – C. Thäle, Osnabrück, Lower Saxony, Germany</i>	2025
<i>Intensive Research Program / Workshop</i>	
Combinatorial Geometries and Geometric Combinatorics <i>Universitat de Barcelona (CRM), Barcelona, Catalonia, Spain</i>	2025
Lattice Theory Workshop <i>The Banff International Research Station (BIRS), Banff, Alberta, Canada</i>	2025
<i>Seminars & Conferences attended</i>	
Algebraic and Enumerative Combinatorics Seminar of the University of Waterloo <i>Karen Yeats, Waterloo, Ontario, Canada</i>	2026
LACIM seminar <i>M. Garcia – S. Giraudo – C. Reutenauer – T. Yu, Montréal, Québec, Canada</i>	2026
Combinatorial Algebra meets Algebraic Combinatorics (CAAC) <i>G. Denham – S. Faridi – S. Kara – H. Thomas – D. Veer, Halifax, Nova Scotia, Canada</i>	2026
Séminaires de mathématiques discrètes <i>Université de Liège, Liège, Liège, Belgique</i>	2025
Nikolaus Conference <i>RWTH Aachen University, Aachen, North Rhine-Westphalia, Germany</i>	2025
Au-delà du Permutoèdre et de l’associaèdre <i>C. Ceballos – V. Pilaud – V. Pons – I. Rasskin, Marseille, Provence-Alpes-Côte d’Azur, France</i>	2025
Closing conference of CGGC <i>Universitat de Barcelona (CRM), Barcelona, Catalonia, Spain</i>	2025
CGGC <i>K. Knauer – A. de Mier – A. Padrol – J. Pfeifle – V. Pilaud, Barcelona, Catalonia, Spain</i>	2025

CGGC25 Participants Seminar	2025
<i>F. Gélinas – L. Jiang – L. Saud, Barcelona, Catalonia, Spain</i>	
Research school of CGGC - Matroid week	2025
<i>Universitat de Barcelona (CRM), Barcelona, Catalonia, Spain</i>	
Research school of CGGC - Polytope week	2025
<i>Universitat de Barcelona (CRM), Barcelona, Catalonia, Spain</i>	
Recollections on polyhedral geometry and (oriented) matroids	2025
<i>Universitat de Barcelona (CRM), Barcelona, Catalonia, Spain</i>	
Seminar on Pure Mathematics	2025
<i>The Hong Kong University of Science & Technology, New Territories, Sai Kung, Hong Kong</i>	
AlCoVe : an Algebraic Combinatorics Virtual Expedition	2025
<i>L. Colmenarejo – M. Gillespie – O. Pechenik – L. Solus – F. Tom – L. Vecchi, Online Conference</i>	
Canadian Discrete and Algorithmic Mathematics Conference (CanaDAM)	2025
<i>Ottawa, Ontario, Canada</i>	
Algebraic and Enumerative Combinatorics Seminar of the University of Waterloo	2025
<i>Karen Yeats, Waterloo, Ontario, Canada</i>	
Combinatorics Seminar of UofT	2025
<i>Vasu Tewari, Toronto, Ontario, Canada</i>	
Winter School and Workshop on Polytopes	2025
<i>M. Juhnke – M. Reitzner Osnabrück – C. Thäle, Lower Saxony, Germany</i>	
LACIM seminar	2025
<i>J. Bastidas – S. Giraudo – C. Reutenauer, Montréal, Québec, Canada</i>	
Combinatorial Algebra meets Algebraic Combinatorics (CAAC)	2025
<i>S. Faridi – S. Kara – R. Orellana – G. Smith – V. Tewari – M. Zabrocki, Toronto, Ontario, Canada</i>	
The Algebraic Combinatorics Workgroup	2022-2025
<i>Nantel Bergeron – Kelvin Chan – Vasu Tewari, The Fields Institute, Toronto, Ontario, Canada</i>	
Applied Algebra Seminar	2022-2025
<i>Lucas Gagnon, York, Ontario, Canada</i>	
Workshop on Computational and Applied Enumerative Geometry	2024
<i>T. Brysiewicz – L. Garcia Puente – N. Hein – J. Lopez Garcia – K. Purbhoo, Toronto, Ontario, Canada</i>	
Combinatorial Algebra meets Algebraic Combinatorics (CAAC)	2024
<i>S. Brauner – J. Levinson – A. Morales – G. Park – F. Saliola – H. Thomas, Montréal, Québec, Canada</i>	
The Graduate Student Seminar UofT	2023
<i>Graduate Student of UofT, Toronto, Ontario, Canada</i>	
Matroids - Combinatorics, Algebra and Geometry Seminar	2023
<i>Ahmed Ashraf, The Fields Institute, Toronto, Ontario, Canada</i>	
Combinatorial Algebra meets Algebraic Combinatorics (CAAC)	2023
<i>Sara Faridi–Selvi Kara –Oliver Pechenik –Matthew Satriano, Waterloo, Ontario, Canada</i>	
The Graduate Students Seminar (UQAM)	2022
<i>Université du Québec à Montréal (UQAM), Montréal, Québec, Canada</i>	
Non-Commutative Algebra, Representation Theory, Special Functions (Conference)	2022
<i>CRM, Montréal, Québec, Canada</i>	
Pan-Quebec colloquium for students of the ISM	2022
<i>Institute of Mathematical Sciences (ISM), Université Laval, Québec, Québec, Canada</i>	

Open Problems in Algebraic Combinatorics (OPAC)	2022
<i>University of Minnesota, Minneapolis, Minnesota, USA</i>	
Graduate Student Conference on Combinatorics, Commutative Algebra and Geometry	2022
<i>University of Minnesota, Minneapolis, Minnesota, USA</i>	
LACIM seminar	2021-2022
<i>LACIM, Montréal, Québec, Canada</i>	

Interests

Combinatorics
Partially Ordered Set (Poset)
Algebraic Combinatorics
Enumerative Combinatorics
Graph Theory
Representation Theory
Algebra
Commutative Algebra
Algebraic Topology